

VLADIVOSTOK, Russian Federation

WMO: 319600

Lat: 43.1201N

Lon: 131.9155E

Elev: 618

StdP: 14.37

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-6.3	-22.4	1.6	-6.0	-18.5	2.0	-3.0	33.2	8.9	29.5	6.2	4.5	0	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	82.6	70.4	79.1	68.9	76.3	67.8	73.7	78.4	72.1	75.9	70.5	74.0	8.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	122.1	75.4	70.9	116.5	74.0	69.4	110.5	72.6	37.8	78.1	36.3	75.5	34.9	74.0	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	29.2	25.3	22.3	DB	-16.3	88.1	8.9	3.9	-22.6	90.9	-27.8	93.2	-32.8	95.4	-39.2	98.2
(5)				WB	-17.2	77.1	8.6	2.1	-23.4	78.6	-28.4	79.8	-33.2	81.0	-39.4	82.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	8.8	15.9	29.3	42.9	51.6	59.2	66.8	69.7	61.4	47.9	29.2	13.7	(6)
(7)		DBStd	21.84	8.15	8.67	7.98	6.22	5.90	5.76	5.36	4.19	5.42	7.64	10.29	8.84	(7)
(8)		HDD50	5037	1278	954	641	228	51	3	0	0	1	131	626	1124	(8)
(9)		HDD65	8867	1743	1374	1106	664	415	190	43	7	130	531	1075	1589	(9)
(10)		CDD50	1938	0	0	0	14	101	278	522	612	345	65	1	0	(10)
(11)		CDD65	291	0	0	0	0	0	15	99	154	23	0	0	0	(11)
(12)		CDH74	1185	0	0	0	4	15	121	435	532	77	1	0	0	(12)
(13)		CDH80	236	0	0	0	1	1	21	104	100	9	0	0	0	(13)
(14)	Wind	WSAvg	9.7	10.7	9.8	10.4	10.5	10.0	9.2	8.8	8.6	8.4	9.8	10.2	9.9	(14)
(15)	Precipitation	PrecAvg	31.9	0.5	0.7	1.0	2.0	3.0	3.8	5.5	6.1	4.6	2.6	1.4	0.8	(15)
(16)		PrecMax	49.6	2.5	2.6	3.7	5.9	7.4	11.0	15.9	12.0	13.9	9.2	5.2	3.1	(16)
(17)		PrecMin	17.8	0.0	0.0	0.0	0.2	0.2	0.4	0.8	0.2	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	8.0	0.5	0.6	0.7	1.2	1.9	2.4	3.3	2.8	3.1	1.9	1.2	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	41.6	53.6	70.6	75.6	82.7	87.6	86.7	80.7	68.4	56.9	41.3	(19)
(20)			MCWB	30.7	33.5	41.0	52.1	58.5	67.9	73.1	71.7	66.6	57.2	48.3	38.8	(20)
(21)		2%	DB	29.9	35.8	47.4	62.8	70.2	77.1	82.6	82.6	75.6	64.5	51.6	36.1	(21)
(22)			MCWB	27.0	31.0	37.2	49.0	55.3	64.5	71.2	70.7	64.3	55.9	45.4	33.0	(22)
(23)		5%	DB	26.2	33.0	43.3	57.3	66.1	73.3	78.8	79.2	72.5	61.8	48.3	32.3	(23)
(24)			MCWB	23.2	29.5	35.1	45.2	53.6	62.6	69.9	70.0	63.4	55.0	43.5	28.9	(24)
(25)		10%	DB	22.6	30.2	40.1	53.3	62.3	68.9	75.2	76.6	69.8	59.3	44.7	28.4	(25)
(26)			MCWB	19.2	26.7	33.4	43.0	52.0	60.9	68.8	69.3	62.1	53.6	40.0	24.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.6	35.3	42.8	53.8	61.0	70.2	76.2	75.9	70.9	62.0	51.0	39.7	(27)
(28)			MCDWB	33.6	38.2	50.4	67.4	72.6	78.8	82.5	81.5	75.3	65.1	53.5	41.0	(28)
(29)		2%	WB	27.2	32.3	38.7	50.1	57.8	66.8	73.7	74.1	68.0	59.6	47.4	33.9	(29)
(30)			MCDWB	29.1	34.5	44.2	60.7	67.0	74.2	80.0	78.3	72.1	62.1	49.9	35.4	(30)
(31)		5%	WB	23.4	29.8	36.6	46.9	55.5	64.4	71.7	72.7	66.2	57.4	44.1	29.1	(31)
(32)			MCDWB	26.2	32.7	41.6	54.3	62.7	70.2	76.5	75.9	69.5	60.2	47.4	32.1	(32)
(33)		10%	WB	19.0	26.7	34.5	44.3	53.6	62.3	69.8	71.4	64.7	55.2	40.4	25.1	(33)
(34)			MCDWB	22.4	30.4	38.8	50.4	59.7	67.1	73.8	74.3	67.9	57.9	44.2	28.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.0	17.8	15.3	16.5	16.5	14.2	12.2	12.3	15.8	16.7	15.7	16.4	(35)
(36)			MCDBR	22.9	22.8	21.1	26.2	26.0	23.9	20.7	18.8	21.2	21.0	21.8	22.3	(36)
(37)			MCWBR	20.3	18.6	13.5	14.2	13.8	13.3	11.1	9.8	11.9	14.2	17.1	20.0	(37)
(38)		5% WB	MCDBR	21.6	20.3	19.2	24.9	22.0	21.0	18.0	13.8	15.4	15.6	19.6	20.8	(38)
(39)			MCWBR	20.3	17.1	13.3	14.4	13.1	12.7	10.7	8.3	9.9	12.4	17.0	19.5	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.321	0.376	0.454	0.546	0.570	0.549	0.519	0.472	0.421	0.430	0.382	0.335	(41)	
(42)		taud	2.194	2.030	1.895	1.742	1.763	1.912	2.064	2.198	2.256	2.080	2.116	2.145	(42)	
(43)		Ebn at Noon	250	251	244	232	231	236	242	250	254	229	221	229	(43)	
(44)		Edh at Noon	30	42	55	69	69	60	51	43	38	40	33	29	(44)	
(45)	All-Sky Solar Radiation	RadAvg	696	1000	1343	1491	1573	1562	1418	1424	1341	1003	692	574	(45)	
(46)		RadStd	44	63	88	134	150	224	187	133	120	72	57	40	(46)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	+4.18	N/A	-2.61	N/A	N/A	N/A	N/A	-158	-79
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page